

AYUSH KUMAR

AI ENGINEER | LLM & FULL STACK DEVELOPER

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PROFESSIONAL SUMMARY

Entry-level AI Engineer and BCA graduate with hands-on experience in LLM applications, RAG pipelines, FastAPI APIs, vector databases and full-stack development. Built AI chatbot workflows using tool calling, Redis/MySQL memory, ChromaDB retrieval and Ollama/Llama 3. Seeking an AI Engineer role to build reliable AI products, automation systems and intelligent backend services.

TECHNICAL SKILLS

AI/LLM: RAG, LangChain, ChromaDB, embeddings, Ollama, Llama 3, nomic-embed-text, prompt engineering, tool calling, LangSmith

Programming: Python, JavaScript, Java, C, C++, PHP

Backend: FastAPI, Node.js, Express.js, REST APIs, JWT authentication

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Angular, Figma

Databases/Tools: MySQL, MongoDB, Redis, SQLite, PostgreSQL basics, Git, GitHub, VS Code, FFmpeg, PyTorch/MPS basics

PROFESSIONAL EXPERIENCE

AI Engineer Intern / Project Development Intern | Paysprint | 18 May 2026 - 17 Aug 2026

- Building an AI assistant/chatbot for business/product support and leave-management workflows using FastAPI, LangChain and RAG.
- Developed RAG pipeline with PDF ingestion, chunking, embeddings, ChromaDB vector search and Ollama/Llama 3 response generation.
- Added Redis short-term memory and MySQL long-term storage to maintain chat sessions, active workflows and user context.
- Designed intent routing and tool-calling logic for policy search, leave request capture, date handling and database actions.
- Used LangSmith-style tracing and evaluation concepts to debug responses, monitor retrieval quality and improve prompt behavior.

Full Stack Developer Intern | North Bihar Power Distribution Company Limited | Jan 2026 - Mar 2026

- Developed a full-stack Employee Management System using the MERN stack.
- Implemented JWT-based authentication and authorization for secure user login and access control.
- Built RESTful APIs using Node.js and Express.js for employee data CRUD operations.
- Designed responsive React.js UI and integrated MongoDB for efficient data handling.

PROJECTS

AI Leave Management & Company Policy Chatbot | Python, FastAPI, LangChain, ChromaDB, Ollama/Llama 3, Redis, MySQL, LangSmith

- Built a conversational AI system that answers company-policy queries through RAG and performs leave-related actions through tools.
- Implemented semantic search using embeddings, vector retrieval and prompt-based response generation.
- Designed active-flow memory so follow-up messages continue the same leave application until required fields are collected.

VideoForge - Local AI Text-to-Video Generator | React, FastAPI, Ollama, PyTorch/MPS, ModelScope, SQLite, FFmpeg, JWT, SSE

- Designed architecture for a privacy-first text-to-video system where prompt input generates video locally without third-party APIs.
- Planned backend pipeline for prompt processing, model execution, video generation, progress streaming and output management.

Frontend Real User Monitoring (RUM) System | JavaScript SDK, Collector API, PostgreSQL/JSONB, React Dashboard

- Designed a frontend monitoring system to capture page views, LCP, INP, CLS, JavaScript errors, resource load timing and device/browser data.
- Proposed lightweight JS SDK, event collector, storage layer and dashboard to help debug real-user frontend performance.

Employee Management System | MERN Stack, JWT, REST APIs, MongoDB

- Built an admin-oriented CRUD web application with authentication, secure employee data handling and responsive React pages.

EDUCATION

BCA - Completed (2023 - 2026): L.N. Mishra Institute of Social Development and Economic Change, Patna (Bihar) - Aryabhatta Knowledge University

Intermediate (2020 - 2022): Global International School, Muzaffarpur (Bihar), CBSE - 74%

Matric (2019 - 2020): Jaintpur Public School, Muzaffarpur (Bihar), CBSE - 64%

ADDITIONAL INFORMATION

Languages: English, Hindi